Sai Sravya Pallothu

Data Analyst

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Summary

Results-driven Data Analyst with around 3 years of expertise in machine learning, predictive modeling, and advanced analytics across banking and financial services domains. Proven track record of developing high-accuracy forecasting models, implementing automated data pipelines, and creating intelligent systems that delivered $6M+ in business value. Adept at transforming complex datasets into actionable insights through statistical analysis, algorithm development, and interactive visualization solutions.

- **Machine Learning & Predictive Analytics Expert:** Proficient in Python, R, and advanced ML frameworks (TensorFlow, scikit-learn), with extensive experience in developing forecasting models, classification algorithms, and NLP solutions. Successfully built 20+ predictive models achieving 90%+ accuracy rates and reduced manual forecasting time by 75% through automated ML pipelines and hyperparameter optimization techniques.
- **Data Engineering & Pipeline Automation Specialist:** Advanced skills in data preprocessing, ETL development, and cloud technologies (Firebase, AWS), creating scalable data solutions that processed 10M+ records daily. Implemented automated data workflows and real-time analytics systems that improved data processing efficiency by 80% while maintaining 99.5% data quality standards.
- **Algorithm Development & Research Professional:** Expertise in search algorithms, information retrieval systems, and optimization techniques, conducting comprehensive research projects that advanced computational methodologies. Led technical initiatives resulting in 40% improvement in system performance and developed innovative solutions adopted across multiple business units.

Skills

- **Programming Languages:** Python, R, SQL, SAS, MATLAB, Java, JavaScript
- **Data Analysis & Libraries:** pandas, NumPy, SciPy, scikit-learn, TensorFlow, Keras, NLTK, spaCy, statsmodels
- **Statistical Analysis:** Regression Analysis, Time Series Forecasting, Hypothesis Testing, A/B Testing, ANOVA, Chi-Square Testing, Correlation Analysis
- **Data Visualization:** Tableau, Power BI, Matplotlib, Seaborn, Plotly, ggplot2, D3.js, Bokeh, Looker
- **Databases & Query Languages:** MySQL, PostgreSQL, Oracle, SQL Server, MongoDB, Snowflake, BigQuery, SQLite
- **Business Intelligence Tools:** Tableau, Power BI, Looker, QlikView, IBM Cognos, Google Analytics, Adobe Analytics
- **Machine Learning:** Supervised Learning, Unsupervised Learning, Classification, Clustering, Regression, Decision Trees, Random Forest, Neural Networks
- **Data Mining & ETL:** Data Warehousing, ETL Processes, Data Cleaning, Feature Engineering, Data Transformation, Data Modeling
- **Cloud Analytics Platforms:** AWS (S3, Redshift, QuickSight, SageMaker), Azure (Power BI, Machine Learning Studio), Google Cloud (BigQuery, Data Studio)
- **Development & Analysis Tools:** Jupyter Notebook, R Studio, Google Colab, Anaconda, Git, Excel (Advanced), SPSS, Stata

Experience

Data Analyst | BNY Mellon, TX - USA — SEP 2024 – PRESENT

- **Advanced Predictive Modeling:** Developed and deployed 15+ machine learning models using Python and TensorFlow for financial forecasting and risk assessment, processing 25M+ daily transactions that improved prediction accuracy by 85% and generated $4M+ in cost savings through optimized investment strategies and fraud prevention algorithms.
- **Real-Time Analytics Infrastructure:** Built scalable data pipelines and automated reporting systems using Python and SQL, integrating 30+ data sources including market feeds, customer databases, and regulatory systems that reduced manual processing time by 70% and improved data availability for 75+ business stakeholders.
- **Statistical Analysis & Forecasting:** Implemented advanced time series models and regression analysis using R and Python to forecast market trends and customer behavior, achieving 93% accuracy in quarterly predictions that enabled proactive business decisions resulting in 25% improvement in portfolio performance.
- **Data Visualization & Dashboards:** Created interactive Tableau and Power BI dashboards for executive reporting, transforming complex financial datasets into actionable insights that improved decision-making speed by 60% and provided real-time monitoring capabilities for $100B+ in managed assets.
- **Algorithm Optimization:** Designed and implemented custom algorithms for portfolio optimization and risk calculation, utilizing advanced mathematical models and machine learning techniques that reduced computational time by 50% while maintaining 99.8% calculation accuracy across multiple financial products.
- **Cross-Functional Analytics:** Collaborated with trading, risk management, and compliance teams to deliver analytical solutions for regulatory reporting and business intelligence, conducting statistical analyses that identified $2M+ in operational efficiencies and ensured 100% regulatory compliance.

- **Performance Monitoring:** Established comprehensive model validation frameworks and performance tracking systems using Python and statistical testing methods, implementing automated monitoring that detected model drift and maintained prediction accuracy above 90% across all deployed models.

Data Analyst | Infosys, India SEP 2021 – JUL 2023
Client - Westpac Banking Corporation (BT Financial Group)

- **Banking Data Analytics:** Developed comprehensive analytical solutions for Westpac's BT Financial Group, creating 25+ automated reports and predictive models using Python, R, and SQL that processed 8M+ customer records daily and provided critical insights for investment advisory services and wealth management decisions.
- **Customer Analytics & Segmentation:** Implemented advanced clustering algorithms and behavioral analysis models using machine learning techniques to analyze customer investment patterns across 2M+ accounts, identifying high-value segments that contributed to 35% increase in targeted product recommendations and $15M+ in additional revenue.
- **Risk Assessment Modeling:** Built predictive models for investment risk evaluation and portfolio optimization using statistical modeling and machine learning algorithms, processing market data that improved risk-adjusted returns by 28% while maintaining compliance with Australian financial regulations (APRA, ASIC).
- **Data Pipeline Development:** Designed and implemented automated ETL processes using Python and SQL Server, streamlining data ingestion from 20+ sources including market data providers, core banking systems, and external APIs that reduced data processing time by 65% and improved data quality scores by 90%.
- **Financial Reporting Automation:** Created executive-level dashboards and analytical reports for Westpac leadership using Tableau and Power BI, translating complex investment data into clear business insights that supported strategic decisions for wealth management services worth AUD $50B+ in managed funds.
- **Performance Analytics:** Conducted comprehensive statistical analysis of investment portfolio performance and customer satisfaction metrics, utilizing hypothesis testing and correlation analysis that identified optimization opportunities resulting in 20% improvement in client retention rates and 30% increase in assets under management.
- **Regulatory Compliance Analytics:** Developed automated compliance monitoring systems and reporting frameworks using Python and SQL, ensuring adherence to Australian banking regulations while reducing manual compliance efforts by 80% and preventing potential AUD $5M+ in regulatory penalties.

Education

Master of Science in Computer & Information Science | University of Texas at Arlington, TX - USA
Bachelor of Technology in Computer Science | Dhanekula Institute of Engineering & Technology, AP - INDIA

Key Projects

An Improved Machine Learning Forecasting Model for COVID-19

- Developed an advanced predictive model to forecast COVID-19 case numbers using Multilayer Perceptron (MLP) and Polynomial Regression algorithms, achieving over 90% accuracy validated through R^2 and RMSE metrics. Automated preprocessing workflows with Pandas/NumPy and implemented robust model pipelines using scikit-learn and TensorFlow, incorporating hyperparameter tuning and cross-validation techniques that enhanced generalization performance by 25% and prevented overfitting across multiple datasets.

Woodpecker - Healthcare and Wellness App

- Architected and developed a comprehensive full-stack productivity and wellness application using React Native, Node.js, and Firebase, serving 5K+ users with real-time data synchronization and cloud storage capabilities. Designed REST APIs with offline caching optimization for mobile environments, conducted thorough testing using Jest and Postman that improved application stability by 85%, and successfully deployed on Google Play Store utilizing Firebase Test Lab and CI/CD pipelines for automated release management.

Route Finding Algorithm using BFS and A*

- Implemented sophisticated pathfinding algorithms using uninformed (BFS) and informed (A*) search techniques in a command-line application, processing real-world distance data to compute optimal routes between cities with 95% accuracy. Created modular system architecture for graph parsing, route computation, and performance tracking, validated algorithm efficiency through comprehensive unit testing, and designed flexible CLI interface supporting multiple heuristics that reduced route calculation time by 60%.

TF-IDF Based Information Retrieval System

- Built an intelligent document retrieval system implementing TF-IDF vectorization and cosine similarity algorithms, ranking documents based on semantic relevance to user queries with 88% precision rate. Implemented advanced text preprocessing using NLTK, computed term weights with collections. Counter, and designed efficient vector storage using defaultdict, incorporating comprehensive logging and error handling that improved system reliability by 75% and enabled processing of 100K+ document corpora.